

Assignment 3 Solutions

1. In the context where f is a function from τ_1 to τ_2 and g is a function from τ_2 to τ_3 , the expression $f \gg g$ is a function from τ_1 to τ_3 .
2. Given that e_1 is a string and e_2 is an int in the context Γ , the expression $e_1.[e_2]$ is a character in that same context.
3.
$$\frac{e_1 \Downarrow \perp \quad e_2 \Downarrow b}{e_1 \text{ xor } e_2 \Downarrow b}$$
4. $\text{fun } f \rightarrow \text{fun } x \rightarrow \text{fun } y \rightarrow f(x, y)$
5. $\text{fun } f \rightarrow \text{fun } x \rightarrow f(f \times x)(f \times x)$
6. g can be of type $'a \rightarrow \text{int}$ (so g can have a polymorphic type). f then has the type $'a \rightarrow \text{int}$. We know that g outputs an integer because the output of g is used in integer addition. We know that f outputs an int because the body of f is the addition of

two integers. Nothing restricts the input of f or g . This can be verified by letting g be $\text{fun } x \Rightarrow 0$. Note. That g is applied to $f\ x$ and $f\ x$ is an `int` does not restrict the input type of g . It only indicates that g can be applied to integers.

7. Suppose x has type τ_1 and y has type τ_2 .

Then $(k\ y)$ has type $('b \Rightarrow \tau_2)$ list.

Because of h 's type, the type of its first argument x must be the same as the type of elements of its second argument $(k\ y)$. So τ_1 must be the same as $'b \Rightarrow \tau_2$.

Finally, this implies the type of $h\ x\ (k\ y)$ is of type $'b \Rightarrow \tau_2$. All together this means the whole expression has type

$$('b \Rightarrow \tau_2) \rightarrow \tau_2 \Rightarrow 'b \Rightarrow \tau_2.$$

And since there were no restrictions on τ_2 ,

We can write the type using a type parameter:
 $('a \rightarrow 'b) \rightarrow 'b \rightarrow 'a \rightarrow 'b$.

This can be verified by taking:

let $h\ x\ l = \text{match } l \text{ with } [] \rightarrow x \mid x :: xs \rightarrow x$
let $k\ x = [\text{fun } y \rightarrow x]$

8. Γ has to have all the variables that are not locally bound. That is: g, y, x, h

g is a function with output type int because it's used in integer addition. There is nothing restricting the input type of g . The output type of g must match that of f because of the use of equality. The type of x is restricted to be the input type of f and the type of y is restricted to be the same as that of x . The input type of f is int because the input is used in integer addition. The output type of f is unrestricted so the same is true of the output type of g .

All together this gives us:

$$\{x:\text{int}, y:\text{int}, h:'a \rightarrow 'b, g:'c \rightarrow \text{int}\} \vdash e:\text{bool}$$

You can also make the types non-polymorphic:

$$\{x:\text{int}, y:\text{int}, h:\text{int} \rightarrow \text{bool}, g:\text{bool} \rightarrow \text{int}\} \vdash e:\text{bool}$$